POLI 3148 Final Project Report

Evidence of Loyalty Signalling - A Costly but Necessary Strategy for Building Trust

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Abstract

In recent years, the political discourse of Hong Kong has undergone a visible "Mainlandization," characterized by the increasing adoption of *guanqiang* (officialese) and frequent references to Central Government initiatives such as "New Quality Productive Forces" and "Patriots Administering Hong Kong." This paper provides an empirical analysis of loyalty signaling, particularly in the speeches of Hong Kong Chief Executives (CEs) from 1997 to 2022, utilizing text mining techniques to quantify the prevalence of *guanqiang* and different classes of loyalty signalling strategies. We also examine the temporal evolution of semantic similarity of CE speeches relative to those of Mainland Chinese leaders. The findings reveal a distinct cyclical pattern: loyalty signaling intensifies following periods of socio-political friction or "trust crises" between Hong Kong and the Central Government. The research suggests that this discursive convergence is not merely a stylistic shift, but a reparative strategy—a tactical move designed to regain political confidence and signal institutional alignment during times of uncertainty.

Keywords: *Loyalty Signalling, Guanqiang, Central-HK Relationship, Text Mining, Cosine Similarity*

1. INTRODUCTION

Rho-class is specially designed for research articles, technical/lab reports, and academic documentation. This guide will help you understand the features and proper use of the template, enabling you to efficiently structure and format your documents with clarity and precision.

This class includes the following files in the rho-class folder:

Additionally, this template includes a sample document `main.tex`, a bibliography file `rho.bib`, as well as figures, tables and code examples to help you get started quickly.

1.1. Literature Review

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2. METHODOLOGY

2.1. Data Collection

The data utilized in this project is organized according to the following directory structure:

- CE's speeches: `Data/Data_Raw/Speeches_and_Articles`
- Government press releases: `Data/Data_Raw/Press_Release`
- Central leaders' HK-related speeches[1]: `Data/Data_Raw/Central/Liason_Office/State_Leader`

The primary dataset consists of a bilingual parallel corpus containing both Traditional and Simplified Chinese official press releases and CE speeches and articles, with Traditional Chinese materials used for analysis. The data spans five administrative terms (July 1997 to March 2026), capturing official discourse from Tung Chee-hwa, Donald Tsang, Leung Chun-ying, Carrie Lam, and John Lee. [2]

2.2. Data Cleaning & Wrangling

The initial data cleaning phase involved adjusting character encoding for the accurate rendering of Traditional Chinese characters, handling missing values, and stripping HTML tags, punctuation, and extraneous whitespace. A strict length threshold was enforced, excluding any documents with fewer than 2,000 characters. Additionally, temporal metadata was standardized by extracting publication years via regular expressions and formatting dates to DD-MM-YYYY, restricting the analytical window strictly to the 1997–2026 period.

Data wrangling procedures include transforming the unstructured corpus into a format suitable for quantitative analysis, and conducting text normalization, stop word removal to prepare the text for subsequent processing.

2.3. Keyword Detection

The `\rhostart {}` command, provides a personalized lettrine for the beginning of the first paragraph as shown in this example.

2.4. Cosine Similarity Analysis

This class includes a customized design for the table of contents, which is disabled by default. To enable it, insert the `\tableofcontents` command before your first section.

3. RESULTS AND FINDINGS

The density of loyalty-signaling keywords in Chief Executives' speeches varied significantly across administrations (ANOVA result: $F(4,2692) = 558.02$, $p < .001$). Tung established a moderate baseline for loyalty rhetoric, which remained relatively stable throughout his tenure, marked by a slight uptick preceding his resignation. Following Tung, loyalty signaling plummeted. For both Tsang and Leung, the median keyword density dropped to zero, indicating that in at least half of their analyzed speeches, loyalty keywords did not appear a single time. Although Figure

[1]Data regarding central leaders' speeches about Hong Kong was accessed from the Liaison Office of the Central People's Government (LOCPG) website: `http://www.locpg.gov.cn/zxzx/zysy.htm`. But the dataset is restrained to the period from 2008 to 2026.

[2]Press release for current administration (Traditional Chinese) can be found at: `https://www.ceo.gov.hk/tc/press.html`; CE speeches: `https://www.ceo.gov.hk/tc/speeches.html`; Archive of past CE's speeches and press releases:`https://www.ceo.gov.hk/tc/links.html`

1 shows a visual upward trend in signaling at the end of Leung's term, statistical analysis demonstrates no significant difference in the overall signaling density between Tsang and Leung.

A dramatic escalation in loyalty signaling occurred during Carrie Lam's administration. The frequency climbed steadily from 2017, reaching its peak in 2020. Her mean signaling density was more than double that of Tung's and nearly five times higher than Leung's. This rhetorical shift reached its absolute triumph under John Lee's administration. Not only is his keyword density the highest, his lowest Coefficient of Variation (CV) further suggests that his high volume of loyalty signaling is not isolated to specific events; it is a standardized, highly consistent, and baked-in component of his everyday executive discourse.

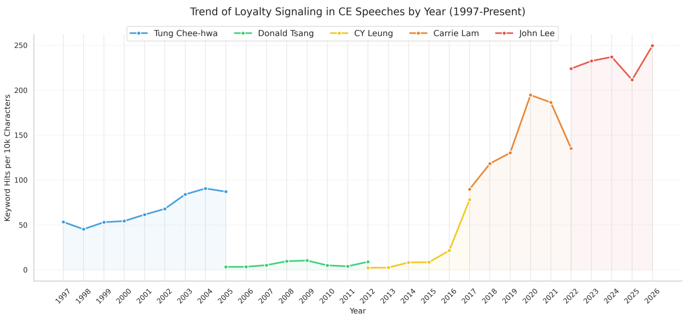

Figure 1. Density of loyalty-signaling keywords across administrations.

Table 1. Statistical Summary and Pairwise Comparison of Loyalty Keyword Density in CE Speeches

		Descriptive Statistics			Post-hoc Comparison (p-adj)			
CE	n	Mean	Median	CV	Tung	Tsang	Leung	Lam
Tung	497	61.15	50.98	0.84	—	—	—	—
Tsang	396	16.20	0.00	2.51	***	—	—	—
Leung	451	29.41	0.00	1.74	***	0.20	—	—
Lam	646	139.93	112.40	0.76	***	***	***	—
Lee	707	228.80	211.60	0.56	***	***	***	***

Note. Overall ANOVA: $F(4, 2692) = 558.02$, $p < .001$.
*** $p < .001$; ** $p < .01$; * $p < .05$.

Chronologically, the year of 2015 serves as a critical inflection point, representing the immediate rhetorical response to the 2014 Umbrella Movement. This shift was preceded by the State Council's June 2014 White Paper, which systemically introduced the doctrine of "Comprehensive Jurisdiction" for the first time.[3] This trajectory of intensified signaling reached its peak in 2020, following the Anti-Extradition Law Amendment Bill Movement . While the promulgation of the National Security Law (NSL) in 2020 was followed by a temporary dip in signaling density, the intensity subsequently surged to an all-time high during John Lee's administration. These patterns strongly support the conclusion that loyalty signaling in the HKSAR is most acute during periods of peak political uncertainty and serves as a reparative mechanism to restore central-local confidence.

While the overall trend for "loyalty signaling" shows a statistically significant increase during the tenures of Carrie Lam and John Lee, this pattern is not uniform across all categories. First of all, there is no rhetorical loyalty signaling throughout all the periods. Secondly, ideological loyalty has constantly remained the most dominant category in terms of absolute volume, reaching a density of 120.4 per 10,000 characters in Lee's administration. But it shall be noted that Tung Chee-hwa also exhibited a high propensity for ideological signaling; as the first Chief Executive, he had to frequently articulate the relationship between the HKSAR and the mainland's socialist system. Thirdly, none of the five CEs demonstrated a significant enthusiasm for "personal loyalty" signaling, indicating that political alignment in the HKSAR is projected as an institutional or systemic commitment rather than an individual allegiance.

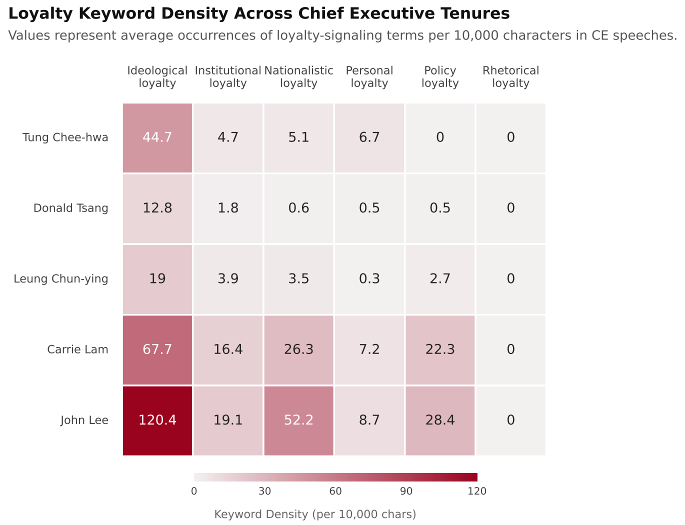

Figure 2. Density of loyalty-signaling keywords by category across administrations.

4. FIGURES

spanishFig.Fig. **??** shows a 3D surface visualization of a bivariate function.

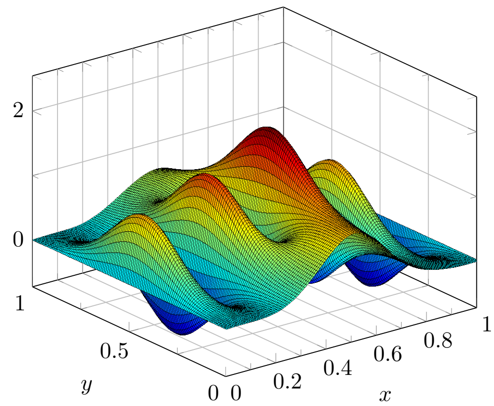

Figure 3. 3D surface obtained from PGFPlots [**PFGPlots**].

5. TABLES

spanishTablaTable **??** shows an example table of some astronomical objects.

Table 2. Astronomical Object Data

Object	Type	Distance (Light Years)
Alpha Centauri	Star system	4.37
Betelgeuse	Red supergiant star	642.5
Andromeda	Spiral galaxy	2.537 million
Earth	Planet	0
Sirius	Binary star system	8.6

Note: The table contains data of some famous celestial objects.

[3] See `https://www.hmo.gov.cn/xwzx/zwyw/202312/t20231208_38591.html` This document signaled Beijing's dissatisfaction with the local administration's perceived inability to manage the precursors to "Occupy Central," emphasizing that Hong Kong's autonomy is a delegated power rather than an inherent right.

The `\tabletext {}` command is used to add notes to tables easily.

6. DOCUMENT DATA

6.1. Front-Matter Footer Block*

A new block called `\docinfo {}`, has been added to the bottom right of the first page, acting as a floating element which stays fixed in place. This replaces the corresponding author block previously located above the abstract in earlier versions.

The `\docinfo {}` block was created to give you greater control over supplementary document data, such as publication dates, licensing information, extended author details, and other front-matter notes. You can fully customize its content to suit your needs.

Additionally, the new `\corres {}` command serves as a flexible alternative to `\thanks {}`. Since modifying the standard `\thanks {}` command can be complex and its positioning is often rigid, `\corres {}` was introduced to offer a more adaptable solution without sacrificing functionality.

spanishNotaNote

You can use `\corres{}` as many times as needed since it automatically ensures that each entry remains correctly associated. For best results, I recommend using text symbols rather than mathematical ones (e.g., `\textasteriskcentered`, `\textdagger`, etc.).

If `\corres {}` is not used, `\docinfo {}` will automatically adjust its position. The implementation was designed to handle any combination of front-matter elements you might need when starting your document. In this example template, you'll find a minimal demonstration that combines both to illustrate their interaction.

6.2. Headers and Footers*

6.2.1. *Headers*

On all pages except the first, the document title and journal appear in the header. Since twoside layout is enabled, its position alternates depending on whether the page is odd or even.

6.2.2. *Footers*

In the first page, six elements display supplementary information for your document:

- `\journalname {}` — Displays the journal name, university, institute, company, or any institutional affiliation.
- `\theday {}` — Displays the current date.
- `\vol {}` — Specifies the volume number of the publication.
- `\no {}` — Specifies the issue number within the volume.
- `\journal {}` — Defines the full journal title.
- `\thepage` — Shows the current page number and the total count.

In all cases, you can freely rearrange or customize the order of these elements by modifying the class file to suit your needs. If any of them are not required, simply omit the corresponding command — the remaining elements will automatically adjust their spacing to remain evenly distributed.

7. RHO CUSTOM PACKAGES

7.1. Rhobabel.sty

This package have all the commands that automatically translate from English to Spanish when this custom package is defined.

By default, this document has its content in English. However, at the beginning of the document you will find a recommendation when writing in Spanish.

You may modify this package if you want to use other language than English or Spanish. This will make easier to translate your document without having to modify the class document.

7.2. Rhoenvs.sty

This package provides custom environments designed to enhance the visual presentation and structure of your document. Key examples include rhoenv, info, and note.

Their style is defined by `rhoenvstyle` — allowing you to customize their appearance according to your document's requirements.

An example using the `rhoenv` environment is shown below.

Custom Title

This is an example of the custom title environment. To add a title type this command `[frametitle=Custom Title]` next to the beginning of this environment (as shown in this example).

The `info` and `note` are environments which have a predefined title and translate their title to Spanish automatically when this language is defined.

8. EQUATIONS

spanishEcuaciónEquation **??** shows the Schrödinger equation as an example.

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left(\frac{*\hbar^2}{2m}(^2 + V(\mathbf{r},t)\right)\Psi(\mathbf{r},t) \tag{1}$$

The `amssymb` and `amsmath` packages were not required, as STIX2 font incorporates mathematical symbols for writing quality equations.

spanishNotaNote

If you would like to change the values that adjust the spacing above and below the equations, change the `\eqskip` value until the preferred spacing is set. The default value is set to 9pt.

9. CODES

9.1. Coding with Minted*

The `minted` package offers customized features for adding codes. In addition, the template is designed to work seamlessly with Fira Code, a monospaced font specially crafted for programming environments.

If your are using a desktop app as TeXstudio, try these steps to make this package work on Windows:

1. Install Python — A stable version (e.g., v.3.11)
2. Open the terminal and type `pip install Pygments`.
3. Update if a newer version is available.
4. Go to TeXStudio settings and change the default compiler to `pdflatex –shell-escape -interaction=nonstopmode \%.tex`.
5. Compile and wait for the result.

Caution

Ensure that **pygments** is properly installed and added to your system's PATH. Otherwise, you may encounter compilation errors. Additionally, enable shell escape when compiling, as it is required for minted to process and highlight code.

In Overleaf, this package is easier to use since it does not require any additional installations or modifications — just add the code as shown in the example. spanishCódigoCode ?? shows a Python example.

```
def greet(name: str) -> None:
    print(f"Hello, {name}.")


def main() -> None:
    greet("world")


if __name__ == "__main__":
    main()
```

spanishCódigoCode 1. Python code example with listings.

You can customize its design changing `\usemintedstyle` command. The different styles offered by the minted package can be preview through this link — `https://pygments.org/styles/`.

9.2. Coding with Listings

Since `minted` requires additional installations and can be complex in some desktop TEX editors, you can use the `listings` package instead, which provides a simpler way to include code.

```
def greet(name: str) -> None:
    print(f"Hello, {name}.")


def main() -> None:
    greet("world")


if __name__ == "__main__":
    main()
```

spanishCódigoCode 2. Python code example with listings.

While `listings` is a simpler and widely supported package for code formatting, `minted` offers a more modern and powerful approach. It leverages the Pygments syntax highlighter to deliver superior coloring, language support, and styling options. One of its key advantages is the ability to easily switch between built-in color styles.

In contrast, `listings` requires manual setup for colors, fonts, and formatting rules. For users who prefer fine-tuned customization, the styling options for `listings` are organized in `rho.cls` file and can be modified at any time to suit individual preferences.

9.3. Inline Code*

As shown in this template, inline code has a custom style for improved visual appeal. To use it, simply add `\inlinecode {}` command and place your code inside the braces.

■ NUMBERLESS SECTION

Declaring a numberless section automatically inserts a square symbol before the title. This unique feature is specific to first-level sections within this class.

As this symbol also affects the table of contents and references titles, users who prefer to disable it may edit the section styling parameters in `rho.cls`.

10. BIBLIOGRAPHY

Bibliography management is handled by `biblatex`, with the default citation and reference style set to IEEE. The `citestyle=numeric-comp` option is enabled, allowing multiple citations to be grouped within a single bracket (e.g., [**davis2023signalflow**, **bell2022codefont**]). The citation format can be customized directly in `rho.cls` to suit any preferred style.

■ FAQ

How do I manage my references?

To manage your references, I recommend using the tool scribbr. You can simply enter the URL or create your own citation, and then export it to LATEX using the options in the three-dot menu.

The generated citation can be copied and pasted into `rho.bib`, the file designated for bibliography management. You may rename this file, but if you do, remember to update the `\addbibresource` command in `rho.cls` under the biblatex section.

spanishNotaNote

Some platforms, such as Google Scholar or scientific journals, provide citations directly in BibTeX format. Therefore, check if there is a "how to cite this document" section to streamline the citation process even further.

If you have any further questions, you can refer to the following page — Bibliography management with biblatex.

What should I do with the example files?

The template includes sample content — such as an example figure, bibliography entries, and the `example.py` script — to demonstrate how the layout and features work. Once you start customizing the document for your own use, feel free to delete all example files and entries you don't need.

How do I place equations easily?

For equations, we have two options: inline or on its own line. For inline equations, simply place a dollar sign ($) at the beginning and end of the equation. However, if you want the equation to be displayed on its own line, you need to use the equation environment.

If you find it challenging to write formulas directly in LATEX, you can use text editors like Word. In the equations menu, you can select LATEX in the conversion section and copy and paste the equation you wrote into one of these environments.

How do I change the paper size?

By default, this class was adapted for a4paper and test it with letterpaper. The following paper sizes are available in LATEX:

- letterpaper (11   8.5 in)
- legalpaper (14   8.5 in)
- executivepaper (10.5   7.25 in)
- a4paper (21   29.7 cm)
- a5paper (21   14.8 cm)
- b5paper (25   17.6 cm)

■ CONTACT ME

Have questions, suggestions, or an idea for a new feature? Found a bug or working on a project you'd like to invite me to?

Feel free to reach out — I'd be happy to help, collaborate, or fix the issue. Your feedback helps me improve my templates!

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GITHUB REPOSITORY

Visit the repository to access the source code, track ongoing development, report issues, and stay up to date with the latest changes.

`https://github.com/MemoJimenez/Rho-class`

SUPPORTING

Enjoyed this document class? Check out tau-class, my very first LaTeX template.

https://es.overleaf.com/latex/templates/tau-class

Any contributions are welcome!

Coffee keeps me awake and helps me create better LaTeX templates. If you wish to support my work, you can do so through PayPal:

`https://www.paypal.me/GuillermoJimeenez`

Enjoy writing with rho-class!